

Technical, economic and environmental assessment and optimization of four hybrid renewable energy models for rural electrification

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ABSTRACT

This work investigates the technical, economic and environmental feasibility of four solar – wind off grid hybrid renewable energy system (HRES) models to provide electrification for Okorobo-Ile town in Andoni Local Government Area of River State, Nigeria using the Hybrid Optimization of Multiple Electric Renewables (HOMER) software. In particular, investigation of the possible inclusion of a fuel cell (FC) system is performed. The four considered models are: pv/wind/battery (PWB); pv/wind/battery/gen-set (PWBG), pv/wind/fuel-cell (PWF) and pv/wind/battery/fuel-cell (PWBf). The best cost-effective configuration among the set of systems were examined for the electricity requirement of 677.75 kWh/day primary load with 99.1 kW peak load. Results obtained showed that the net present cost (NPC) are \$615,664.95, \$679,348.17, \$778,834.22 and \$3,534,850.54 respectively for the PWB, PWBG, PWBf and PWF. The cost of energy (COE) was lowest for the PWB with a value of \$0.158 and highest for the PWF with a value of \$0.964. The renewable options—PWBf and PWF have higher long-term costs but offer cleaner emissions. In contrast, options with the Diesel-Powered Generator is cost-effective but has a high environmental impact in terms of greenhouse gas emissions and noise pollution. These emissions include 3,758 kg/yr CO₂, 23.7 kg/yr CO and a total of 32.67 kg/yr of unburned hydrocarbons, sulfur dioxide, particulate matter and nitrogen oxides. Based on the results, a stand - alone HRES that consist of 166 kW PV panels, 3 wind turbines 29 batteries and 123 kW converter is found to be the best configuration for the village, as it leads to minimum net present cost (NPC) and COE. The PWB system offers the best choice for the community by balancing financial considerations with sustainability which is crucial when making energy system choices. Results also show that while hydrogen, FC system and the electrolyzer can be used together with or without batteries, inclusion of the FC system resulted in the high NPC due to their high cost of investment.

1. Introduction

The main application of the stand-alone renewable energy system is in the rural areas where utility grid lines are expensive to install owing to the terrain or other environmental concerns [1]. Even in the absence of these constraints, building new transmission lines is expensive. Power

generation from wind and photovoltaic (PV) technologies are among the most promising avenues in the renewable energy sector. Additionally, due to their rapid technological advancements and inherent benefits, fuel cell (FC) systems are also demonstrating significant potential when integrated into hybrid renewable energy systems such as flexible modular structure, zero or low emissions (of pollutant gases) and high

Abbreviations: W, Watt; V, Volt; kW, kiloWatt; kg, kilogram; m, meters; kg/yr, kilogram per year; CO, carbon monoxide; HRES, hybrid renewable energy system; DC, direct current; Homer, Hybrid Optimization of Multiple Electric Renewables; FC, fuel cell; PWB, pv/wind/battery; PWBG, pv/wind/battery/gen-set; PWF, pv/wind/fuel-cell; PWBf, pv/wind/battery/fuel-cell; kWh/day, kiloWatt hour per day; PV, photovoltaic; LGA, Local government area; NPC, net present cost; NASA, National Aeronautics and Space Administration; TED, total daily energy demand; PWB, PV module/wind turbine/rechargeable battery and converter; PWBG, PV module/wind turbine/rechargeable battery/diesel generator and converter; PWBf, PV module/wind turbine/rechargeable battery/fuel-cell and converter; PWF, PV module/wind turbine/fuel-cell and converter; COE, cost of energy.

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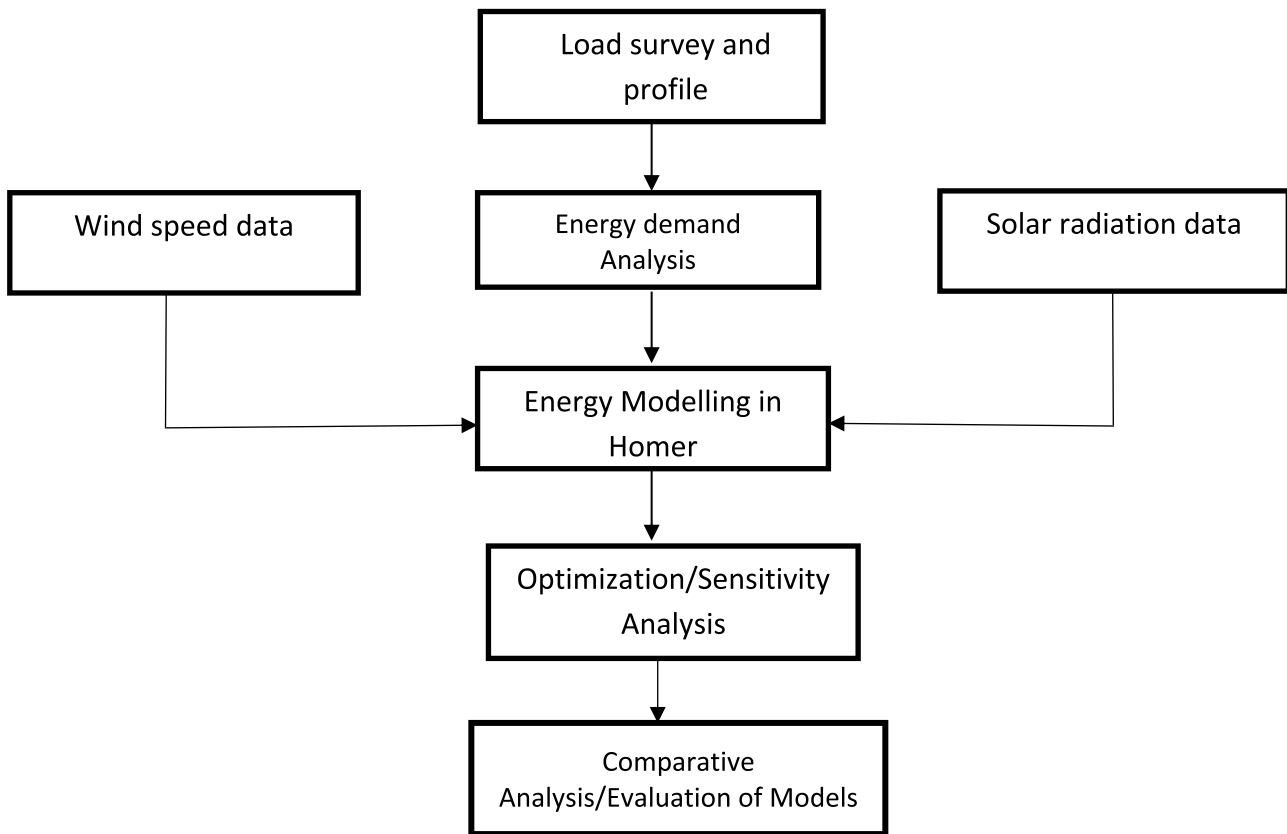


Fig. 1. Research framework.

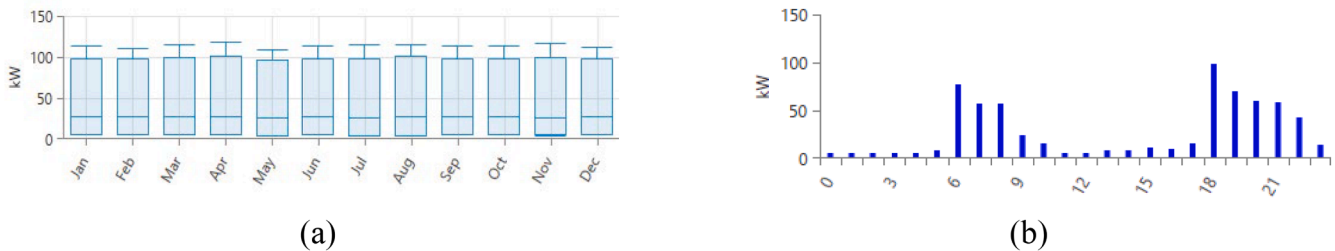
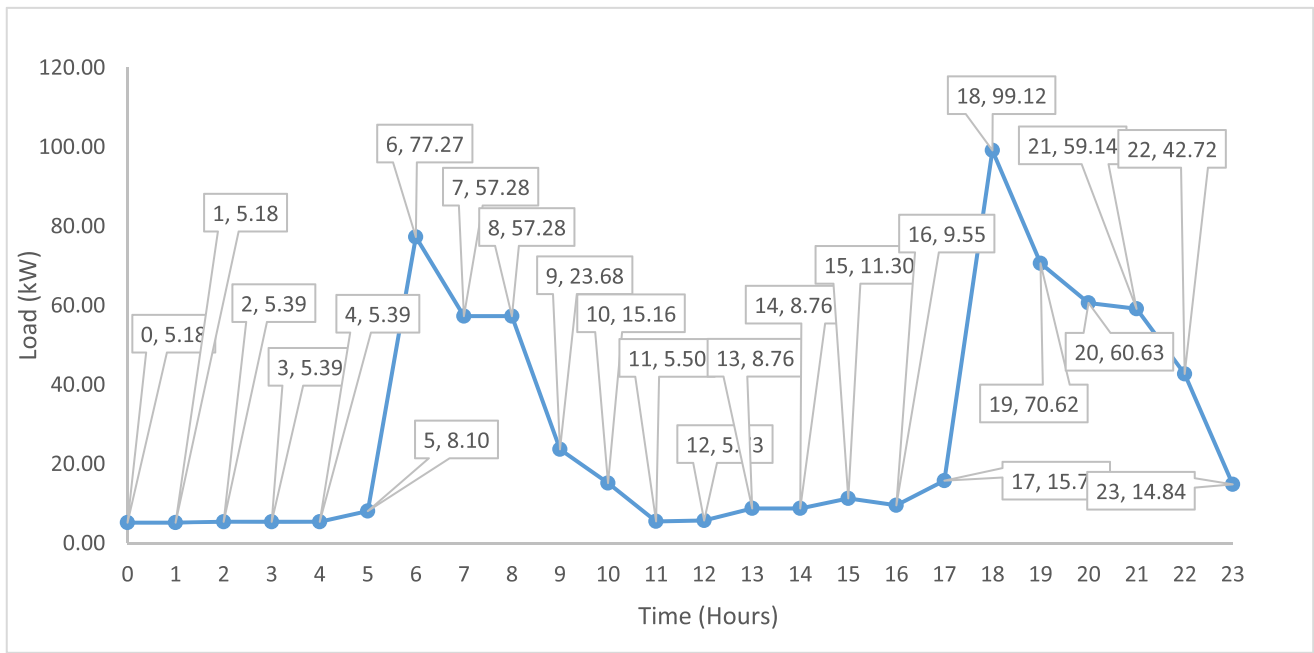


Fig. 2. Load profile of Okorobo-Ile town in Homer (a) Seasonal (b) Daily.

efficiency. Inclusion of electric vehicles (EV's) in HRES is also on the rise [2]. Residents of rural communities spend more than their inadequate earnings on fuel for their basic electrical energy needs. Interestingly, these communities have an abundance of sunlight and wind that could be harnessed to meet their basic electricity requirements. Simple utilization of PV panels and wind turbines could capture sufficient energy to provide this essential power. This could be the very first steps in eradicating starvation in rural communities where little or nothing is usually done to provide basic needs for human life. Owing to the lack of these technologies at affordable prices, there is hindrance in investments in such communities. There is need to solve this high price problems using low budget solutions. One possible solution to this problem is the use of hybrid renewable energy systems. Imagine rural communities in Andoni LGA of Rivers State where there is need to cook food, pump water, etc. There is therefore need to improve the design of HRES in order to minimize cost and make it easier to support thousands of rural residents who are constantly starving. That is why this work aims at the viability of a HRES for the region. A major concern of hybrid renewable energy systems is their stochastic nature – solar energy is only available during the day time and the load pattern is often not correlated with the wind

speed. These fluctuations and uncertainties in HRES can be alleviated by including energy storage and also nonrenewable energy generations. The design of a stand-alone hybrid system is more complex and costly due to the need to integrate multiple power sources with different performance characteristics. This contrasts with a grid-connected system, which is less complicated as it supplements the existing utility grid [1]. Owing to these, there are a large number of parameters and variables that requires optimization in order to achieve the cost-effective and technical design objectives. Keeping in mind the above facts, design and implementation of a cost-effective hybrid renewable energy system for electrification is the best solution. In their comprehensive review [3], discussed hybrid architectures that combine wind, solar, biomass, and geothermal energy for off-grid applications. They propose performance indicators for techno-economic assessment which provides valuable insights for optimizing rural electrification models. This work therefore aims at a techno-economic feasibility study and analysis of a hybrid power system that meets the electricity needs of Okorobo-Ile town in Andoni Local Government Area of Rivers State, Nigeria. Hybrid renewable energy systems (HRES) have been extensively studied for their potential to provide sustainable and reliable power, particularly in



(a)



(b)

Fig. 3. (a) Detailed load profile of Okorobo-Ile town [20] (b) Aerial view of the village (Google map).

remote and rural areas of Nigeria. Nnadi et al. [4] demonstrated the feasibility of hybrid solar-wind systems for electrification in South-Eastern Nigeria, highlighting their benefits for remote areas. Olatomiwa et al. [5] conducted a techno-economic analysis of hybrid PV–diesel–battery and PV–wind–diesel–battery systems for mobile base transceiver stations, emphasizing their cost-effectiveness and reliability for rural development. Olatunji et al. [6] optimized the performance of HRES for a rural healthcare center in Kano, underscoring the importance of optimization in enhancing system performance. Oladigbolu et al. [7] performed a comparative study and sensitivity analysis of standalone hybrid energy systems for rural healthcare facilities, highlighting the need for tailored solutions. Olatomiwa [8] assessed the optimal configuration of hybrid renewable power supply for rural healthcare facilities, demonstrating their economic viability. Olatomiwa and Mekhilef [9] evaluated the techno-economic feasibility of HRES for rural health centers, advocating for their adoption to improve healthcare delivery. Ajayi et al. [10] designed an optimal hybrid renewable energy system for isolated-grid communities in Northern Nigeria, providing a

comprehensive design framework. Amoo [11] evaluated the wind energy potential of two sites in South-West Nigeria, contributing valuable data for wind energy integration. Diemuodeke et al. [12] performed a multi-criteria assessment of HRES for Nigeria's coastline communities, emphasizing the importance of considering multiple factors in system design. Oyedepo et al. [13] studied thermospheric winds over Abuja during the solar minimum period, providing insights into atmospheric conditions that could affect renewable energy systems. Ikejamba and Schuur [14] developed a multi-step approach for locating solar and wind parks in South-Eastern Nigeria to maximize population coverage. Oza-veshe and Jen [15] analyzed the energy cost of hybrid systems and diesel generators for powering base transceiver stations, highlighting the economic advantages of hybrid systems. Jumare et al. [16] conducted an environmental life cycle assessment of grid-integrated HRES in Northern Nigeria, providing insights into their environmental impacts. Mohammed et al. [17] conducted a technical and economic analysis of a hybrid microgrid power system using HOMER Legacy software, providing a detailed assessment of system performance and cost-effectiveness.

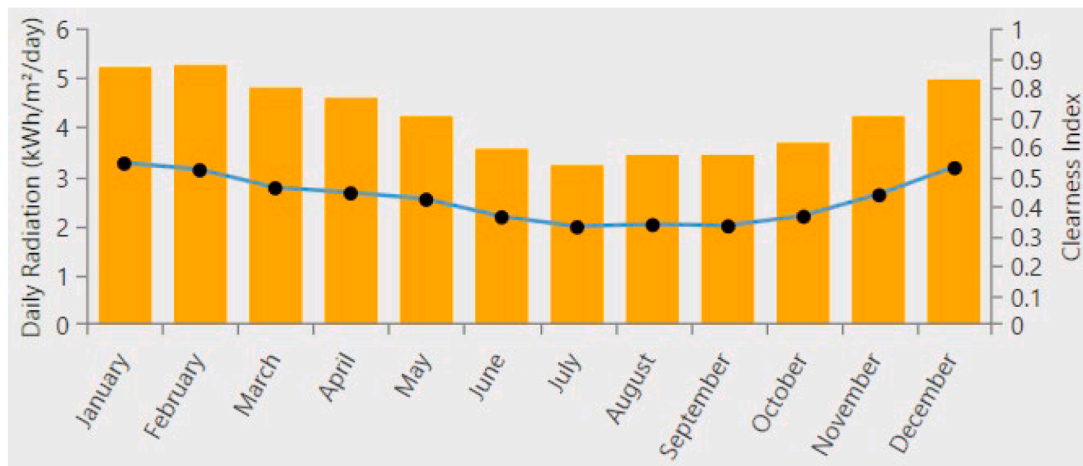


Fig. 4. Solar radiation profile with clearness index from NASA.

Ohijeagbon et al. [18] designed an optimal hybrid renewable energy system for sustainable power supply to isolated-grid communities in North Central Nigeria, demonstrating practical implementation strategies. Babatunde et al. [19] performed a techno-economic assessment of photovoltaic-diesel generator-battery systems for base transceiver stations, emphasizing their economic benefits. Ukoima et al. [20] analyzed a solar hybrid electricity generation system for a rural community in River State, Nigeria, demonstrating its effectiveness in providing sustainable power. Dioha and Kumar [21] estimated the technical potential of rooftop solar PV for urban residential buildings in Nigeria, using computational methods to highlight the significant potential for decarbonizing the electricity sector. Azimoh and Mbohwa [22] explored optimized solutions for increasing electricity access with mini-grid technology, showing the viability of renewable energy resources in different regions of Nigeria. Ole [23] examined the role of financial support under international climate change regimes in combating electricity poverty through off-grid renewable electricity, highlighting the financial barriers and potential solutions. Soremekun [24] detailed the development of an 85kWp solar minigrid in Gbamugbamu, Ogun State, showcasing a practical implementation of renewable energy solutions in a rural community. Collectively, these studies underscore the potential of HRES to address the energy needs of remote and rural areas in Nigeria, offering a sustainable, reliable, and cost-effective solution for electrification. The literature review clearly indicates that for regions without electricity access or those experiencing significant power outages, hybrid renewable energy systems are the most viable and cost-effective solution. However, despite the extensive research on HRES in Nigeria, there remains a significant gap in the exploration of integrating renewable energy sources with hydrogen or battery storage specifically for rural electrification in Nigeria. To the best of the authors' knowledge, none of the existing studies have investigated the potential of combining renewable energy with hydrogen or battery storage in this region. This study is the first to explore such combinations for the specified geographical zone, making it novel in its approach.

In light of this, the present work seeks to evaluate four different renewable energy systems to determine which can most economically fulfill the energy requirements of Okorobo-Ile town in Andoni LGA. To achieve this objective, four models will be designed and analyzed using Homer Pro software. This study contributes the following:

1. Propose an economically feasible stand-alone hybrid renewable energy system that provides electrification to a rural village in Nigeria.
2. Optimally size system components in the proposed model.
3. Study the viability of the proposed system.

The analyzed models are generally made up of a combination of

solar, wind, battery, fuel cell and diesel generator. Homer software, which was developed by the National Renewable Energy Laboratory United States of America, is used for simulation. Owing to its renowned status for giving accurate optimization results, homer software was used in this study. A plethora of geographical information (abundant renewable energy resources and an almost accurate component cost) is contained within the software [25–28]. The software provides cost optimization of designed hybrid systems by ranking results obtained based on the net present cost (NPC) method [29–32]. The rest of this paper is organized as follows: Section 2 gives an overview of the methodology, Section 3 presents the results obtained, Section 4 discusses the results. The conclusion is given in Section 5.

2. Methodology

The steps taken in this work to reach the desired objectives are illustrated in Fig. 1.

2.1. Description of Okorobo-Ile town and load profile

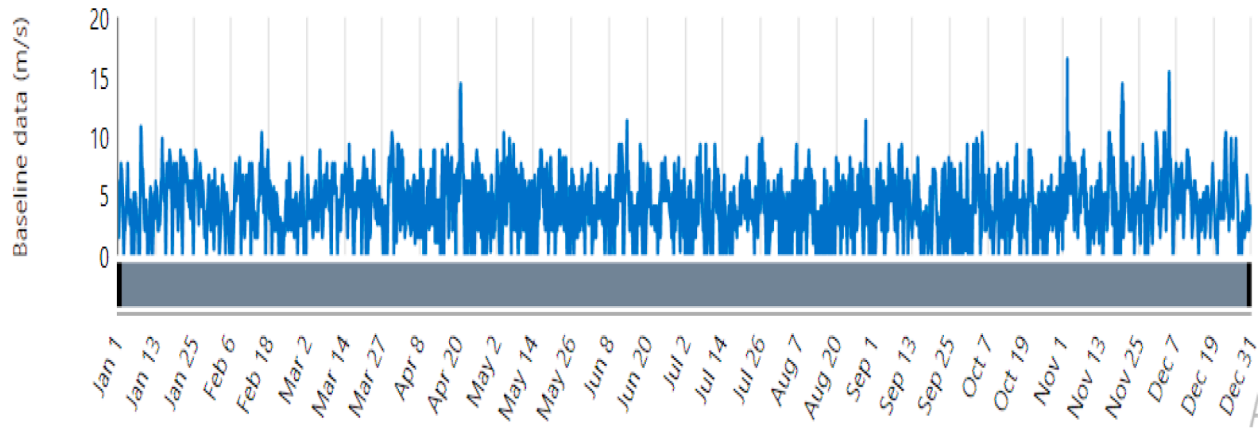
Okorobo-Ile town is a remote settlement in South – South, Nigeria. It is located at the boundary between Rivers and Akwa-Ibom State. The village is home to approximately 6700 residents, encompassing around 600 households, and includes community centers such as schools, churches, and a town hall. The majority of the villagers depart their homes in the morning and return in the evening. The community's total daily energy demand (TED) is 656.36 kWh, with a peak load of 99.12 kW and a total daily load of 678 kW [20]. This is shown in Fig. 2. Fig. 3 represents a detailed load profile and shows that the peak usage of power occurs in the evening around 6pm. A random variability of 5 % was applied to both the day to day and time step variability in homer.

2.2. Resource data

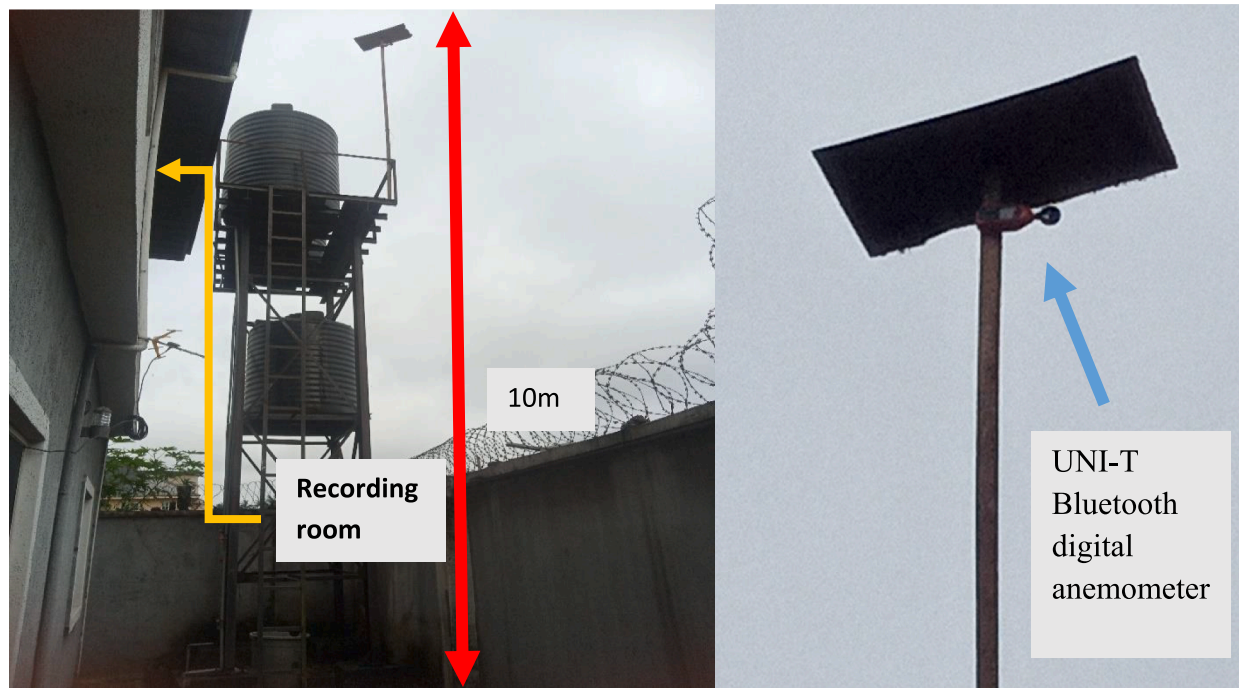
The site solar data was obtained from NASA solar energy radiation database in HOMER as shown in Fig. 4. The wind speed data was obtained from hourly measurement from January 2020 – January 2023) using the UNI-T Bluetooth digital anemometer installed at a 10 m height. The UNI - T Bluetooth enables data transfer from the device installed at the measurement height to the mobile app installed in a mobile phone. Fig. 5a shows a plot of the site wind speed.

2.3. Proposed models of the hybrid system

Four models are being designed and analyzed. The considered hybrid system models consists basically of wind turbine, an array of PVs and



(a)



(b)

Fig. 5. (a) Wind speed profile from physical measurement (b) Set up for wind speed measurement.

battery storage. Fig. 6 shows the various configurations and set up in homer. The four models are:

- i. **PWB**: This model consist of wind turbines, PV modules, batteries and converters.
- ii. **PWBG**: This model consist of wind turbines, PV modules, gen set, batteries and converters.
- iii. **PWBF**: This model consist of wind turbines, PV modules, fuel cell sub system, batteries and converters.
- iv. **PWF**: This model consist of wind turbines, PV modules, fuel cell sub system, and converters.

2.4. Sizing and optimization

Homer utilizes two distinct optimization algorithms. The first, a grid search algorithm, performs simulations of all potential system configurations as determined by the search space. The more recent optimizer in Homer uses a unique algorithm that doesn't require derivatives to

pinpoint the most cost-effective system. Homer then gives a configuration list sorted by their net present cost which is also referred to as the life-cycle cost. This can be used to evaluate different system design options. The Homer optimization software streamlines the identification of the most cost-effective system configuration, enhancing the efficiency of the selection process across various sensitivity scenarios. It facilitates the examination and juxtaposition of diverse system designs, including those that are entirely reliant on renewable resources. Optimization tables are particularly useful for contrasting different system layouts, detailing specific architectural elements like battery count, converter dimensions, or photovoltaic capacity. Additionally, these tables offer financial insights, presenting figures for energy costs, net present cost (NPC), operational expenses, and upfront investment, alongside metrics like the proportion of renewable energy within the system [31–33]. Homer uses equations 1–8 to compute the NPC and COE. Table 1 shows the inputs in homer for sizing and optimization. Fig. 7 gives an overview of the optimization process in Homer.

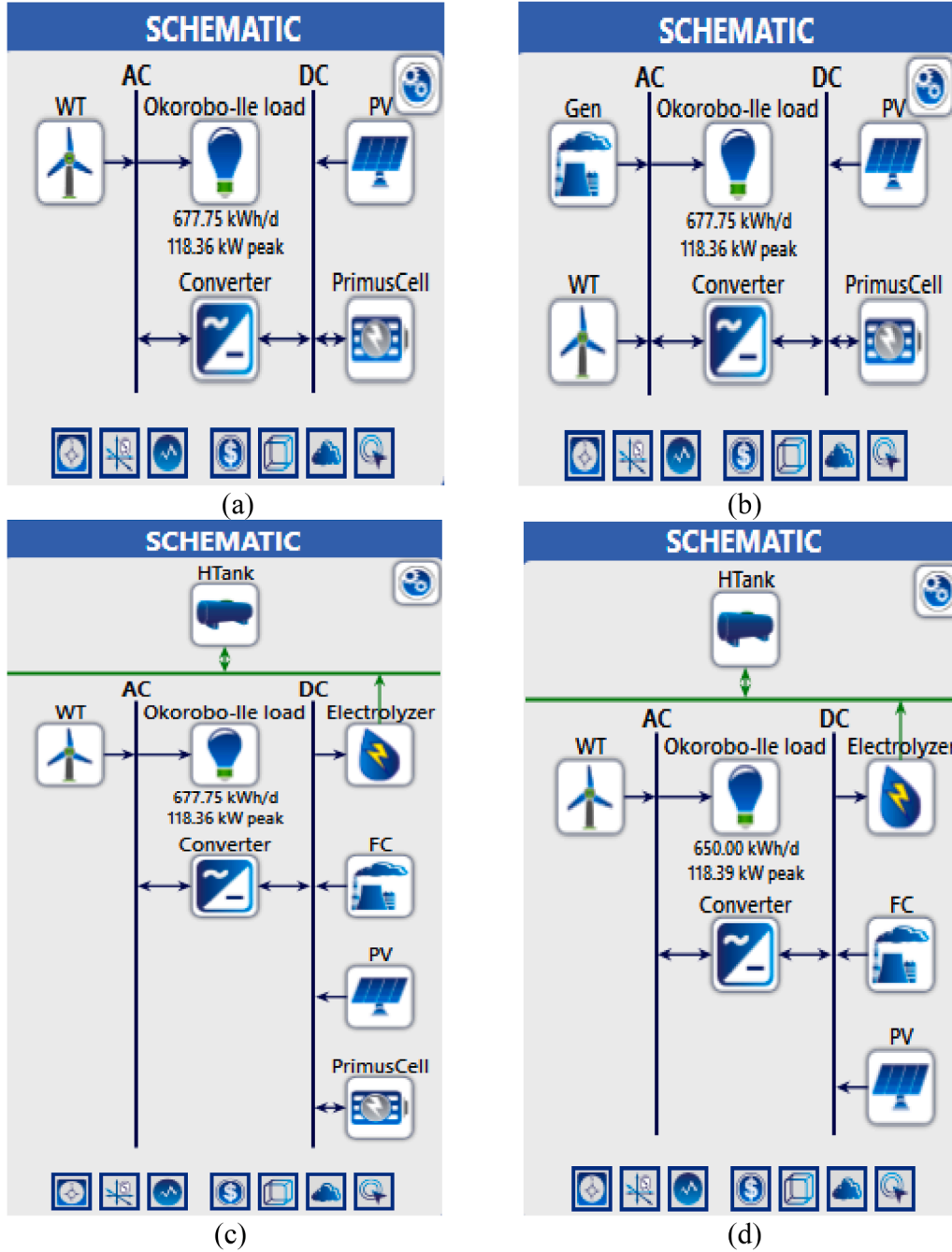


Fig. 6. Proposed models: (a) PWB, (b) PWBG, (c) PWBF, and (d) PWF.

$$NPC = \sum \text{capital cost} + \text{replacement cost} + \text{maintanance cost} - \text{salvage cost} \quad (1)$$

$$COE = \text{total annualized cost (\$/yr)} / \text{total electric load served (kWh/yr)} \quad (2)$$

$$\text{Capital cost} = \sum_{i=1}^n C_{i,\text{capital}} \quad (3)$$

$C_{i,\text{capital}}$ is the capital cost of the component (i) (e.g., PV panels, wind turbines, batteries, fuel cell, generators)

$$\text{Replacement cost} = \sum_{i=1}^n \frac{C_{i,\text{replacement}}}{(1+r)^{t_i}} \quad (4)$$

$C_{i,\text{replacement}}$ is the replacement cost of the component (i), r is the discount rate, and t_i is the time at which replacement occurs.

$$\text{Maintenance cost} = \sum_{i=1}^n C_{i,\text{maintenance}} \times T \quad (5)$$

$C_{i,\text{maintenance}}$ is the annual maintenance cost of component (i), and T is the project life time.

$$\text{Salvage cost} = \sum_{i=1}^n \frac{C_{i,\text{capital}} \times \left(\frac{T_{i,\text{remaining}}}{T_i} \right)}{(1+r)^T} \quad (6)$$

$T_{i,\text{remaining}}$ is the remaining life of component (i) at the end of the project and T_i is the total useful life of component (i).

Table 1
Homer inputs for the various models.

	PV array	Wind turbine	Converter	Battery	Gen set	Hydrogen tank	Electrolyzer	Fuel cell
Unit	kW	kW	kW	kW	kW	kW	kW	kW
Size	1	25	1	72	1	1	1	1
Capital cost (\$)	1200	50000	300	4000	400	130	1400	1800
Replacement cost (\$)	1000	50000	300	3500	400	120	1300	1700
O&M cost (\$/yr)	10	500	0	8	0.03	0.2	20	0.2
Life time	25yrs	20yrs	15yrs	10yrs	15,000hrs	25	15	40,000hrs
PV/WIND/BATTERY	Size considered Quantities considered	Optimizer 0,1,2,3	- Optimizer	Optimizer	Optimizer			
PV/WIND/BATTERY/GENSET	Size considered Quantities considered	Optimizer	3	Optimizer	Optimizer	Optimizer		
PV/WIND/BATTERY/FUEL CELL	Size considered Quantities considered	200,220,240		100,120,125,130				
		2		20,25,30	40,50,80,100			
PV/WIND/FUEL CELL	Size considered Quantities considered	Optimizer	2	Optimizer		500,600,700,800,900	100,150,200	100,150,200

* Discount rate = 30%. Projects life time = 25 years.

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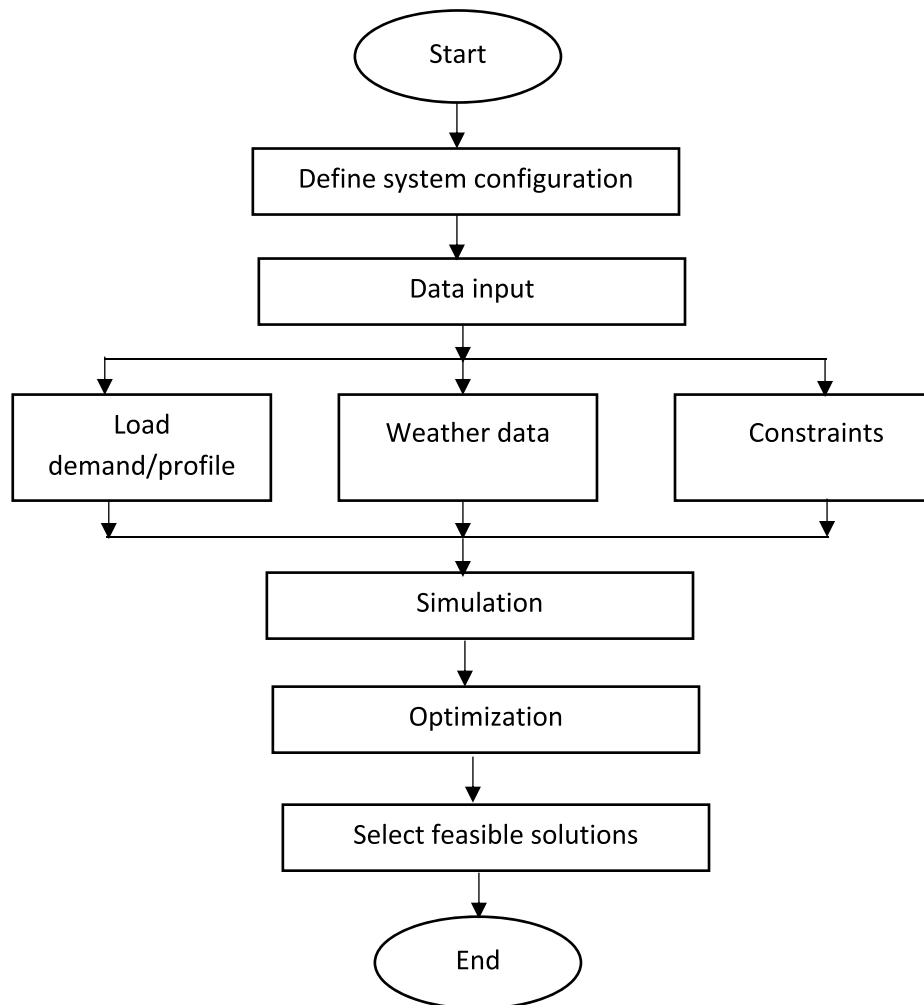
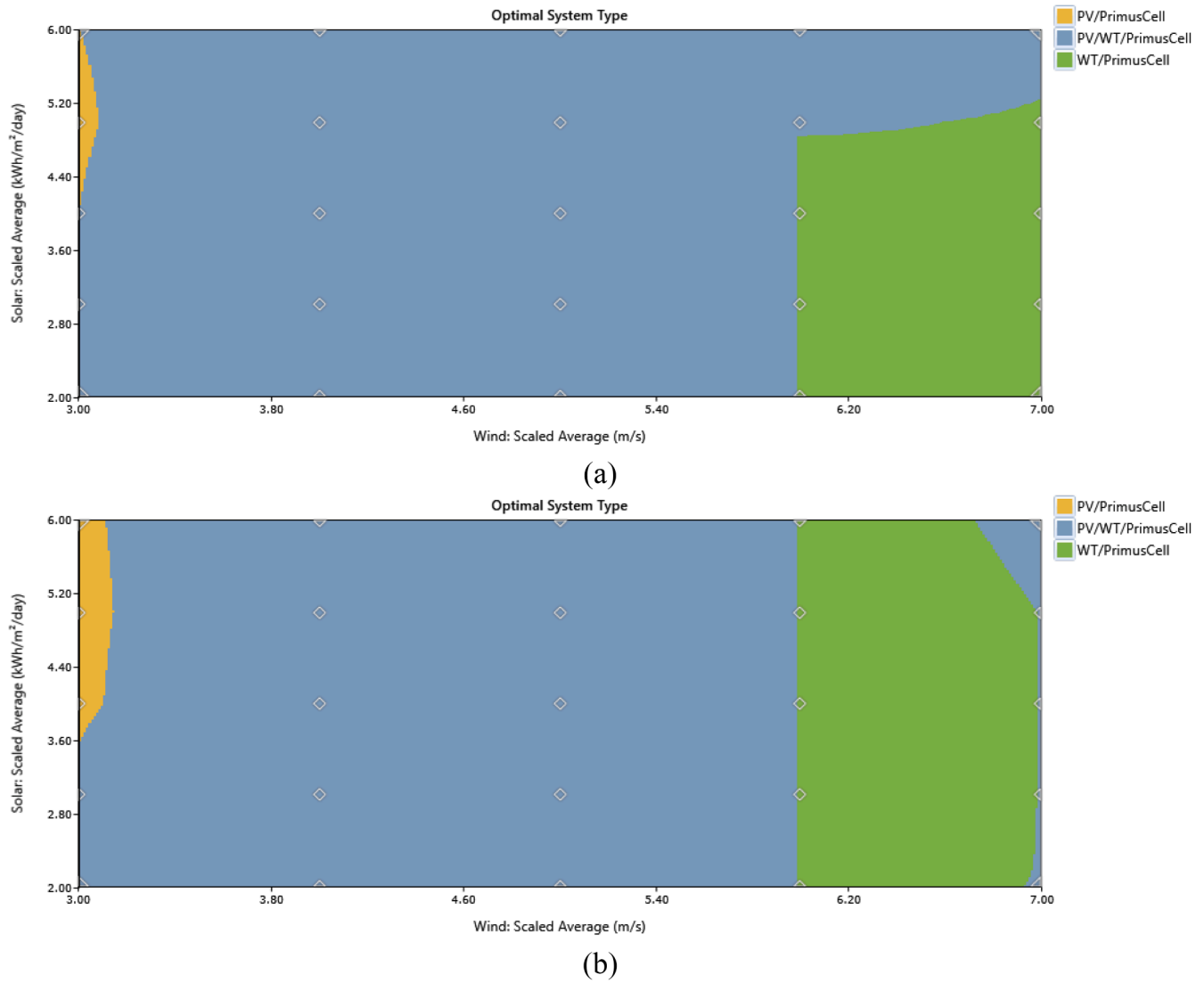


Fig. 7. Homer optimization process flowchart.

Table 2

Cost summary for PWB.

Component	Capital	Replacement	O&M	Salvage	Total
Batteries	\$116,000.00	\$47,027.53	\$3654.56	\$29,099.46	\$137,582.62
Wind turbine	\$150,000.00	\$69,498.81	\$23,628.60	\$43,004.13	\$200,123.28
PV	\$198,889.67	\$0.00	\$26,108.24	\$0.00	\$224,997.92
System Converter	\$36,928.22	\$20,738.29	\$0.00	\$4705.38	\$52,961.13
System	\$501,817.89	\$137,264.63	\$53,391.39	\$76,808.98	\$615,664.95

**Fig. 8.** Sensitivity graph for PWB system (a) with battery capital cost of 1 (b) with battery capital cost of 0.5.**Table 3**

Cost summary for PWBG.

Component	Capital	Replacement	O&M	Fuel	Salvage	Total
Battery	\$100,000.00	\$40,540.97	\$3150.48	\$0.00	\$25,085.74	\$118,605.71
Autosize Genset	\$56,000.00	\$0.00	\$2712.56	\$40,703.24	\$19,943.72	\$79,472.08
Wind turbine	\$150,000.00	\$69,498.81	\$23,628.60	\$0.00	\$43,004.13	\$200,123.28
PV	\$199,000.74	\$0.00	\$26,122.82	\$0.00	\$0.00	\$225,123.56
System Converter	\$39,063.55	\$21,937.46	\$0.00	\$0.00	\$4977.46	\$56,023.54
System	\$544,064.29	\$131,977.25	\$55,614.46	\$40,703.24	\$93,011.06	\$679,348.17

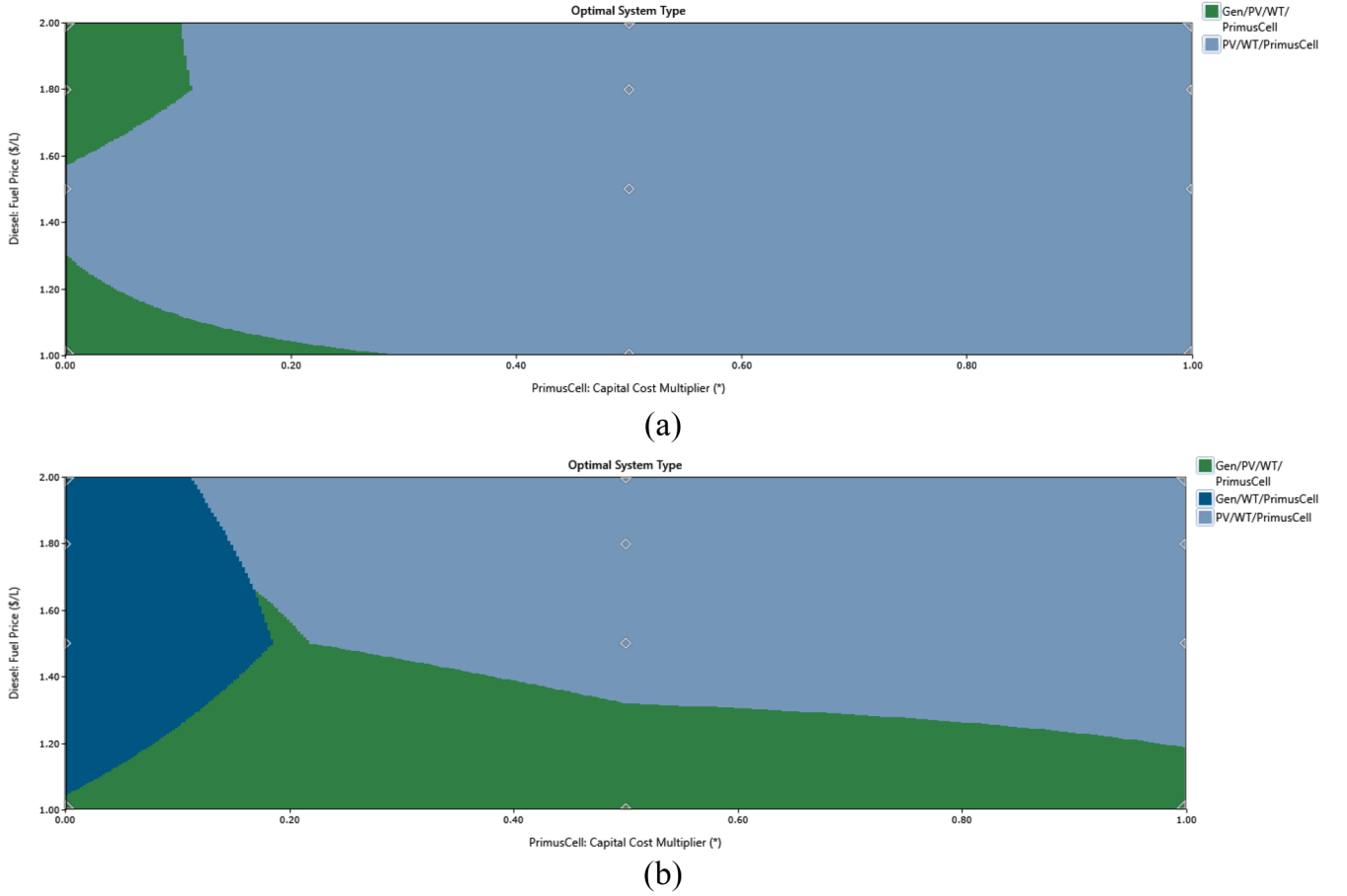


Fig. 9. Sensitivity graph for PWBG system (a) at hub height of 50 m (b) at hub height of 60 m.

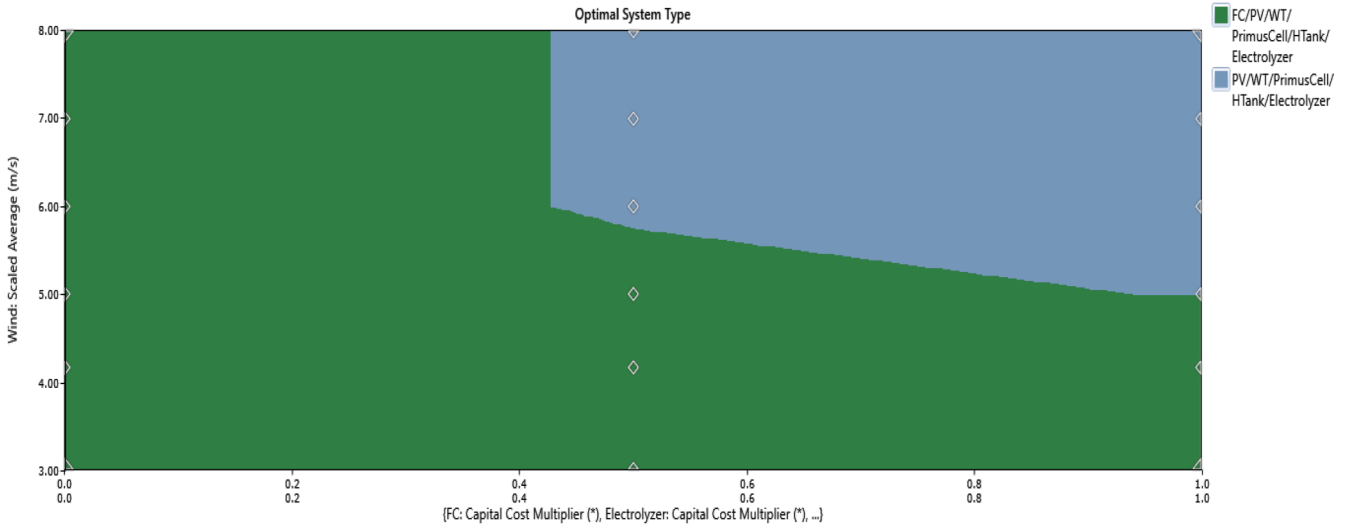


Fig. 10. Sensitivity graph for PWBFB.

$$\text{Total annualized cost} = \text{NPC} \times \text{CRF} \quad (7)$$

CRF is the capital recovery factor and is defined as:

$$\text{CRF} = \frac{r(1+r)^T}{(1+r)^T - 1} \quad (8)$$

Table 1 presents the various costs and component lifetimes. Our assumptions align with studies on hybrid renewable energy systems.

Photovoltaic (PV) panels have a lifetime of 25 years, based on industry standards. Wind turbines operate for 20 years, consistent with typical operational lifetimes. Batteries and converters are assumed to last 10 and 15 years respectively reflecting the average life expectancy of commonly used battery and converter technologies. These components are replaced at the end of their lifetimes.

Table 4

Cost summary for PWBF.

Component	Capital Replacement	O&M	Fuel	Salvage	Total	
Battery	\$120,000.00	\$48,649.17	\$3780.58	\$0.00	\$30,102.89	\$142,326.85
Wind turbine	\$100,000.00	\$46,332.54	\$15,752.40	\$0.00	\$28,669.42	\$133,415.52
Fuel cell	\$36,000.00	\$0.00	\$8947.36	\$0.00	\$11,843.34	\$33,104.02
Electrolyzer	\$70,000.00	\$36,502.95	\$15,752.40	\$0.00	\$8282.28	\$113,973.07
PV array	\$240,000.00	\$0.00	\$31,504.79	\$0.00	\$0.00	\$271,504.79
Hydrogen tank	\$19,500.00	\$0.00	\$472.57	\$0.00	\$0.00	\$19,972.57
Converter	\$45,000.00	\$25,271.27	\$0.00	\$0.00	\$5733.88	\$64,537.39
System	\$630,500.00	\$156,755.93	\$76,210.10	\$0.00	\$84,631.81	\$778,834.22

Table 5

Cost summary for PWF.

Component	Capital Replacement	O&M	Fuel	Salvage	Total	
Wind turbine	\$100,000.00	\$46,332.54	\$15,752.40	\$0.00	\$28,669.42	\$133,415.52
Fuel cell	\$270,000.00	\$292,311.10	\$1897,376.19	\$0.00	\$47,824.18	\$2411,863.12
Electrolyzer	\$210,000.00	\$109,508.85	\$47,257.19	\$0.00	\$24,846.83	\$341,919.21
PV array	\$446,827.60	\$0.00	\$58,655.05	\$0.00	\$0.00	\$505,482.65
Hydrogen tank	\$91,000.00	\$0.00	\$2205.34	\$0.00	\$0.00	\$93,205.34
Converter	\$34,141.64	\$19,173.39	\$0.00	\$0.00	\$4350.32	\$48,964.72
System	\$1151,969.24	\$467,325.89	\$2021,246.16	\$0.00	\$105,690.75	\$3534,850.54

3. Results

3.1. Technical, economic and environmental analysis and optimization

3.1.1. PWB system

For this model, the winning solution consists of 166 kW PV panels (Generic flat plate PV), 3 wind turbines (Eocycle EO25 Class III), 29 batteries (20kW-72 kWh Primus Power Energy Cell) and 123 kW converter. The PV array generates 204,703 kWh/yr while the wind turbine generates 236,121 kWh/yr serving a primary load of 247,236 kW/yr. The system report indicates an excess electricity generation of 127,646 kWh per year. This surplus power is not a drawback as it can offset losses in the distribution network and accommodate potential load growth in the near future [33–35]. An annual conservative growth rate of 2 % in energy consumption was assumed. This assumption was based on historical data and projected trends in energy consumption for Okorobo-Ile town, as detailed in a study using the RETScreen software [20]. This excess electricity provides a sufficient buffer to accommodate this anticipated growth, ensuring reliability and efficiency.

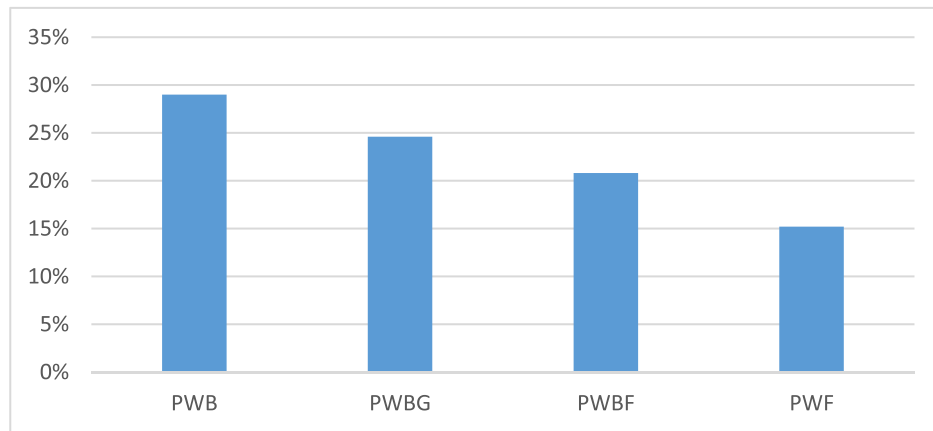
This excess electricity can also be exported to the National grid and can support the expansion of the micro grid to nearby communities, providing them with reliable and sustainable energy. With a renewable energy fraction of 100 %, the system does not contribute to greenhouse gas emissions. The autonomy of the system is 73.9hr (3 days) with cost of energy (COE) of \$0.158. Table 2 shows the cost summary for this model with NPC as \$615,664.95. The PV modules have the highest initial capital cost which is approximately 39.6 % of the total initial cost. Wind turbines follow making up about 29.9 % of the initial cost. Batteries and the system converter contribute 23.1 % and 7.4 % respectively. Replacement costs are significant for wind turbines (\$69,498.81) and batteries (\$47,027.53), indicating a need for future investment. O&M costs are highest for PV modules (\$26,108.24) and wind turbines (\$23,628.60), reflecting ongoing maintenance needs. The total system cost is \$615,664.95, with PV modules contributing the most at 36.5 %, followed by wind turbines at 32.5 %. Batteries and the system converter add 22.4 % and 8.6 % respectively. The considered sensitivity variables for this model are the battery capital cost multiplier (0, 0.5, 1), wind speed (3, 4, 5, 6, 7) and solar irradiance (2, 3, 4, 5, 6). By varying these parameters for the model, 85,783 simulations were performed out of which 32,154 were feasible. The results as seen in Fig. 8 show that with reduced battery costs, homer recommends PV – battery for increased solar irradiance and wind – battery system when the wind speed is high.

3.1.2. PWBG system

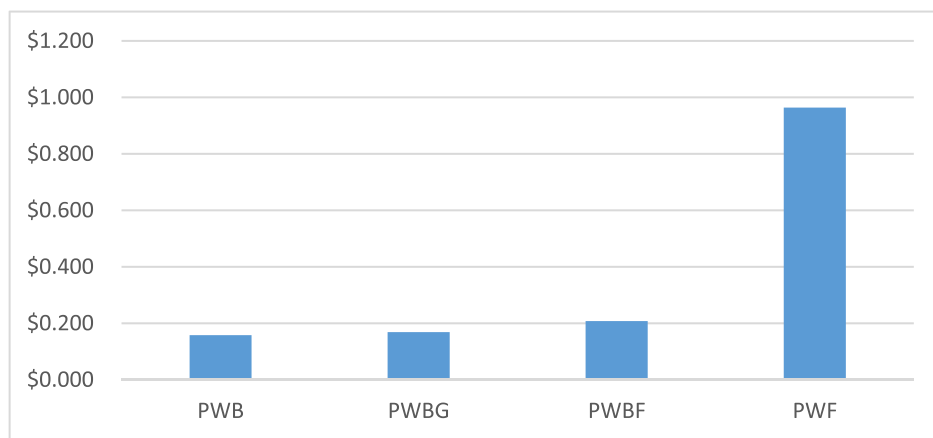
For this model, the winning solution consists of 140 kW PV panels, 3 wind turbines, 29 batteries, 133 kW converter and 140 kW gen set. The PV array generates 173,044 kWh/yr, the wind turbine generates 236,121 kWh/yr. Homer recommends a 140 kW diesel generator that contributes 7850 kWh/yr to the total power generated serving the primary load. The system report shows that there is an excess electricity of 133,092 kWh/yr. As a result of the inclusion of the gen set, there is emission of greenhouse gases. These emissions include 3758 kg/yr CO₂, 23.7 kg/yr CO and a total of 32.67 kg/yr of unburned hydrocarbons, sulfur dioxide, particulate matter and nitrogen oxides. The autonomy of the system is 73.9hr (3 days) with cost of energy (COE) of \$0.169. Table 3 shows the cost summary for this model with NPC as \$659,966.80. Similar to the PWB model, here, the PV modules have the highest initial capital cost approximately 36.6 % of the total initial cost. Wind turbines follow making up about 27.6 % of the initial cost. Batteries and the system converter contribute 18.4 % and 7.2 % respectively, while the genset adds \$56,000.00 (10.3 %). The genset incurs fuel costs of \$40,703.24, highlighting its operational expense. For sensitivity analysis of this model, the considered variables are the wind turbine hub height (30, 40, 50, 60), diesel price (1, 1.5, 1.8, 2) and battery capital cost multiplier (0, 0.5, 1). With these sensitivity parameters, there were 29,932 solutions out of which 23,984 were feasible. At a hub height of 30 m and 40 m, a PV-Wind-battery system is recommended. At 50 m hub height, the PV-Wind-battery system occupies a large part of the sensitivity space as seen in Fig. 9. A PV-Wind-Gen-Battery system is recommended for reduced cost of battery and diesel. At 60 m hub height, a wind – battery – gen system is recommended only when the cost of battery is reduced.

3.1.3. PWBF system

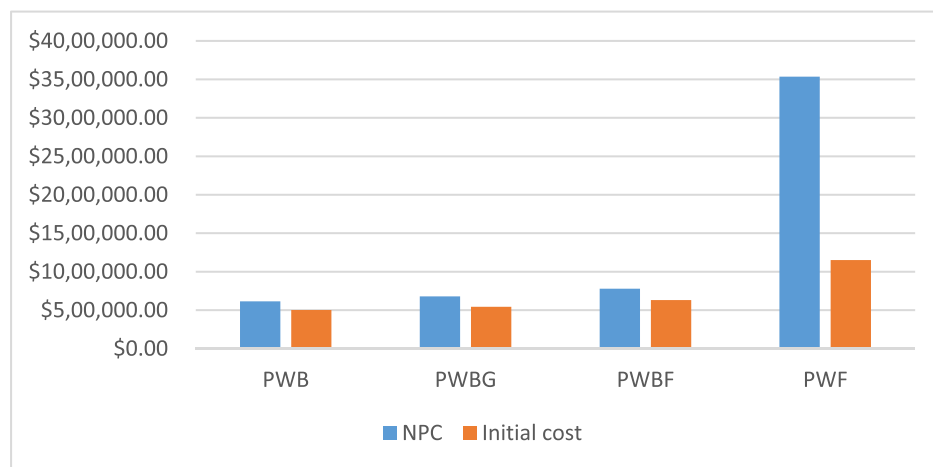
The optimal system type graph as seen in Fig. 10 shows the conditions under which it makes economic sense to invest in fuel cell. Investment in fuel cell makes sense when their cost is low and high wind speed for this model. The finding is that fuel cells are not competitive at high prices, but if their price was cut in half they would compete under certain circumstances. The winning solution for this model consist of 220 kW PV array, 2 wind turbines, 20 kW fuel cell, 25 batteries, 40 kW electrolyzer, 40 kg hydrogen tank and 120 kW converter. The cost summary of this model is seen in Table 4. The system has a COE of \$0.18, NPC of \$681,950.84 with battery autonomy of 79.8 hrs. The PV array has the highest initial capital cost approximately 38.1 % of the total



(a)



(b)



(c)

Fig. 11. Cost comparison between all models: (a)% excess electricity (b) COE (c) Initial cost and NPC.

initial cost. Wind turbines follow making up about 15.9 % of the initial cost. Batteries and the electrolyzer contribute 19.0 % and 11.1 % respectively, while the fuel cell adds 5.7 %. The hydrogen tank and converter add 3.1 % and 7.1 % respectively. Replacement costs are significant for batteries (\$48,649.17) and the electrolyzer (\$36,502.95), indicating a need for future investment. In this model, 58,320 solutions were simulated of which 47,655 were feasible.

3.1.4. PWF system

The winning solution in his model consists of 372 kW PV array, 2 wind turbines, 150 kW fuel cell, 150 kW electrolyzer, 700 kg hydrogen tank and 114 kW converter. Surplus electricity from solar and wind sources is directed to an electrolyzer, producing hydrogen that is then stored in a tank. This hydrogen is later used by a fuel cell to generate electrical power. However, the overall efficiency of this hydrogen

Table 6

Economic and environmental implications of the various models.

System Model	Short-Term Cost (\$)	*Long-Term Cost (\$)	Environmental Impact
PWB (PV/Wind/Battery)	615,664.95.	1732,063.70	No greenhouse gas emissions
PWBG (PV/Wind/Battery/Gen set)	659,966.80.	1857,115.21	Greenhouse gas emissions due to diesel generator
PWBF (PV/Wind/Battery/Fuel cell)	681,950.84	1919,894.18	Lower emissions compared to diesel generator
PWF (PV/Wind/Fuel cell)	\$3.53M	9939,842.91	Lower emissions compared to diesel generator
Diesel-Powered Generator	79,472.08	223,825.33	High greenhouse gas emissions
Grid Power	-	-	Depends on the source of electricity

Was computed using an inflation rate of 0.4 and life span of 20 years.

Long term cost = short term cost $(1 + \text{inflation rate})^{\text{Life span}}$.

storage cycle is below 50 %, as reported by Homer Energy in 2023. Due to the significant energy loss during storage, the output from photovoltaic panels and wind turbines needs to surpass the actual electricity demand to compensate [36–38]. The system has a COE Of \$0.964, NPC of \$3.53 M with battery autonomy of 79.8 hrs. The PV array has the highest initial capital cost approximately 38.8 % of the total initial cost followed by the fuel cell with about 23.4 % of the initial cost. The electrolyzer and wind turbines contribute 18.2 % and 8.7 % respectively, while the hydrogen tank and converter add 7.9 % and 3.0 % respectively. O&M costs are highest for the fuel cell (\$1897,376.19) and PV array (\$58,655.05), reflecting ongoing maintenance needs. The total system cost is \$3534,850.54, with the fuel cell contributing the most at 68.2 %, followed by the PV array at 14.3 %. The electrolyzer and wind turbines add 9.7 % and 3.8 % respectively, while the hydrogen tank and converter contribute 2.6 % and 1.4 % respectively. In this model, 58,320 solutions were simulated of which 1620 were feasible. Table 5 shows the cost summary for this model

4. Discussion

The complexity of power generation systems that rely on renewable energy is often heightened by the issue of storing energy. While batteries provide a solution for short-term energy storage needs within these systems, converting electricity into hydrogen serves as a strategy for sustainable development and facilitates long-term energy storage [33]. Renewable energy resources, when used for hydrogen production in a fuel cell system results in the production of fuel that is environmentally friendly. Given the aforementioned results from the simulation of the PWB and PWF systems, using the batteries together with electrolyzer and hydrogen tank is conceivable. It is also feasible to use both batteries

and hydrogen. However, the inclusion of hydrogen increases the total cost of the system due to increased components cost. A combination of the photovoltaic array, wind turbines, battery bank, and the fuel cell system increases the cost of energy and net present cost when compared with the PWB and PWBG systems. This combination uses a reduced number of batteries, a smaller hydrogen tank, and a low-power electrolyzer. In the PWF system where there are no batteries and gen set, there is a significant difference between the COE and NPC when compared with the PWB and PWBG systems. Considering hydrogen's status as a clean energy source with a substantial energy density, it presents a preferable alternative to batteries or other storage devices for electricity. The inclusion of fuel cells in the PWBF and PWF models also aligns with the findings in [39] on the benefits of integrating advanced technologies for enhanced performance. However, the high costs associated with these models underscore the need for careful financial planning and potential subsidies to make such systems viable in developing regions. In the light of this, implementation of fuel cell technology in Nigeria would incur high expenses, rendering it an economically impractical option. The PWB system on the other hand has the least COE and NPC of \$0.158 and \$615,664.95 respectively as seen in Fig. 11 even though it has the most percentage of excess electricity (29 %). This excess electricity is beneficial for reason stated in the previous section. Hence, the PWB system is the most cost-effective and economic configuration. This confirms report in [20,51] that a PV/wind/battery hybrid renewable energy system is the optimal configuration for the region. Table 6 shows a comparison of the short-term and long-term costs of various energy systems over a 20-year period. A significant increase due to inflation can be observed. The renewable options—PWBF (PV/Wind/Battery/Fuel cell) and PWF (PV/Wind/Fuel cell)—have higher long-term costs but offer cleaner emissions. In contrast, the Diesel-Powered Generator remains cost-effective initially but has a high environmental impact in terms of CO₂ emissions and noise pollution. These emissions consists of 3758 kg/yr CO₂, 23.7 kg/yr CO and a total of 32.67 kg/yr of unburned hydrocarbons, sulfur dioxide, particulate matter and nitrogen oxides. These emissions have adverse effects on both human health and the environment. Mitigating them through cleaner energy sources and improved combustion technologies is crucial for a sustainable future. In this regard, the PWB system offers the best choice for the community by balancing financial considerations with sustainability which is crucial when making energy system choices. In terms of practical implications, the PWB system reduces dependence on fossil fuels, mitigating its environmental impact. Policymakers and stakeholders should prioritize investments in the PWB systems to maximize the economic benefits and sustainability. For the PWBF and PWF systems which have higher costs, their environmental benefits make them suitable for areas with stringent environmental regulations or where long-term sustainability is a priority. Government subsidies and incentives can help offset these costs. The use of fuel cells can also enhance energy security by diversifying the energy mix and reducing vulnerability to fuel price fluctuations. Table 7 presents a comparison of this work to similar studies. The cost of energy (COE) in this study ranges

Table 7

.Comparing present study with some recent PV/wind HRES studies.

Ref	Location	Load Demand	COE (\$/kWh)	Total Cost (\$)	Rural/Urban	Storage Type	No. of models	Environmental Analysis
[40]	Cameroon	165.44 kWh	-	-	Urban	Battery	1	X
[41]	Bangladesh	350 kWh	0.29	175,938	Rural	Battery	1	✓
[42]	Ethiopia	2.4 MW (avg)	0.126	6897,300	Rural	Battery	1	✓
[43]	India	73.6 MW (peak)	0.05744	180,200,000	Urban	Battery	1	✓
[44]	Cameroon	1.5 MW	0.214	1200,000	Rural	Battery	1	✓
[45]	Afghanistan	500 kWh/day	0.25	200,000	Rural	Battery	1	✓
[46]	Nigeria	1.2 MW	0.18	950,000	Rural	Battery	1	✓
[47]	Kenya	800 kWh/day	0.22	150,000	Rural	Battery	1	✓
[48]	Nepal	600 kWh/day	0.20	120,000	Rural	Battery	1	✓
[49]	Tanzania	1.0 MW	0.19	850,000	Rural	Battery	1	✓
[50]	Uganda	900 kWh/day	0.21	140,000	Rural	Battery	1	✓
Ours	Nigeria	677.75 kWh/day	0.158 – 0.964	615,665 – 3534,851	Rural	Battery/Fuel cell	4	✓

from \$0.158 for the PV/Wind/Battery (PWB) system to \$0.964 for the PV/Wind/Fuel-Cell (PWF) system, which is within the range observed in other studies. Also, this study's total costs varies from \$615,664.95 for the PWB system to \$3534,850.54 for the PWF system, reflecting the higher initial investment costs associated with fuel cell systems. Furthermore, this study's emphasis on environmental benefits, particularly the cleaner emissions of the PWB and PWF systems, aligns with the common focus on sustainability in other studies. The use of battery storage in the PWB system as the most cost-effective option is also a trend seen in other research. This study fills a significant gap by exploring the integration of fuel cell and battery storage for rural electrification in Nigeria, making it a new contribution to the field.

5. Conclusion

This paper investigated the technical, economic and environmental assessment and optimization of four hybrid renewable energy models - Photovoltaic/wind turbine/rechargeable battery, Photovoltaic/wind turbine/rechargeable battery/genet, Photovoltaic/wind turbine/rechargeable battery/fuel cell and Photovoltaic/wind turbine/fuel cell that meets the electric power needs of a rural community in Rivers State. The PWF model, with its reliance on wind, solar, and hydrogen fuel cells, offers the most robust and environmentally friendly energy generation and storage solution. It has the highest initial capital costs, particularly for PV modules (38.8 %) and the fuel cell subsystem (23.4 %), as well as the highest replacement costs for the fuel cell (\$292,311.10) and electrolyzer (\$109,508.85). The PWF model also incurs significant O&M costs, especially for the fuel cell (\$1897,376.19), but these are offset by its low environmental impact and high salvage values for the fuel cell (\$47,824.18) and wind turbines (\$28,669.42). The findings indicate that while the electrolyzer, fuel cell (FC), and hydrogen tank can be integrated with batteries, incorporating the FC system led to the highest Net Present Cost (NPC) due to the elevated cost of components. Also, the inclusion of generator sets in the PWBG is not environmentally friendly due to the emission of greenhouse gases. These emissions include 3758 kg/yr CO₂, 23.7 kg/yr CO and a total of 32.67 kg/yr of unburned hydrocarbons, sulfur dioxide, particulate matter and nitrogen oxides. The findings further indicate that the PWB system which comprise of a combination of 166 kW photovoltaic panels, 3 wind turbines, 29 batteries, and a 123 kW converter constitutes the optimal setup, resulting in the lowest net present cost (NPC) of \$615,664.95 and a cost of energy (COE) of \$0.158/kWh for the area. The system's design, with an excess electricity generation of 127,646 kWh per year, provides a sufficient buffer to accommodate the anticipated growth, ensuring reliability and efficiency. The results of this study have significant practical implications for rural electrification projects in regions with similar geographical and socio-economic conditions. The PWB system, with its lower NPC and COE, offers a viable solution for providing reliable and sustainable electricity to rural communities. A comprehensive lifecycle cost analysis to better understand the long-term economic and environmental impacts of each model can be conducted as a future study. Lastly, policy makers and stakeholders should engage in the training of local technicians on HRE systems and engage communities in the planning process. These are crucial for the success and sustainability of these projects.

CRedit authorship contribution statement

Kelvin Nkalo Ukoima: Writing – review & editing, Writing – original draft, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Okoro Ogbonnaya Inya:** Writing – review & editing, Visualization, Validation, Supervision. **Akuru Udochukwu Bola:** Writing – review & editing, Visualization, Funding acquisition. **Davidson Innocent Ewean:** Writing – review & editing, Visualization, Funding acquisition.

Conflict of interest

We confirm that this work is original and has not been published elsewhere, nor is it currently under consideration for publication elsewhere. The authors have no conflicts of interest to disclose and have not received any funding

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